VIMAL JYOTHI
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Assignment Cover Page

Name of the student :- NAMITH.K.P	
PRN VML22CS127	Admission No. 7965
Subject Name :- Data Compression Techniques	Subject Code:- CST446
Assignment Title/No : 1	
Name of the faculty: Aiswarya M.R	
Assignment Submitted on 27/02/26	

Late submission rules : Max mark will reduced to 50% for 1-5 working day's delay, no mark will be awarded thereafter.

I am hereby confirming that this assignment is my own and I haven't adopted any unfair means in any steps of its preparation to enhance my performance in this assignment.

Date : 27/02/26

Sign with Name
NAMITH.K.P

Assignment subdivision	Maximum Mark	Marks awarded	Remarks
A			
B			
C			

Feed back/suggestions :

Name and sign of the faculty

10. Statistical Methods are important in Data Compression because they assign shorter Codewords to symbols that occur more frequently and longer Codewords to less frequent symbols. This reduce the average Number of bits per symbol, thereby improving compression efficiency.

The efficiency of a compression technique is measured by the average Code length, which depends on symbol probabilities. The objective of statistical compression is to design codes whose average length is as close as possible on the entropy of the source.

Therefore Statistical Methods are important because they,

* Reduce redundancy in data.
* Minimize average code length.
* Increase Compression efficiency
* Ensure unique and accurate decoding.

→ Two Statistical Coding Methods are,

→ Huffman Coding.

Huffman Coding is an optimal prefix Coding technique that generates variable-length code words based on symbol probabilities.

* Algorithm steps.

1. List all symbols in decreasing order of probability
2. Combine the two symbols with the lowest probabilities.
3. Assign binary digits 0 and 1 to the branches.
4. Repeat the process untill only one combined symbol remains.
5. Determine the codewords by tracing the path from the root to each leaf node.

* Features.

* Produces prefix codes
* more frequent symbols receive shorter codewords
* Minimizes the average code length for a given probablity distribution.
* provides lossless Compression.

→ Arithmetic Coding.

Arithmetic Coding is a Statistical Compression Method that represents an entire sequence of symbols as a single fractional number b/w 0 and 1

- Working Principle :

* The interval [0,1] is divided according to cumlative symbol Probabilities.
* For each symbol processed the interval is reduced proportionally
* After encoding all symbols, any number within the final interval uniquely represents the entire message.

- Features.

* Does not assign separate Codewords to individual symbols.
* Achieves Compression efficiency closer to entropy
* Suitable for sources with skewed probability distribution.